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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of NITTE (Deemed to be University)
Second Semester B.Tech (CBCS) Degree Examinations
May 2025

EC1002-2 – APPLIED DIGITAL LOGIC DESIGN
(For AC, EC, EE, VL, AD, AM, CB, CC, CS, IS, RI)

Duration: 3 Hours

Max. Marks: 100

Note:

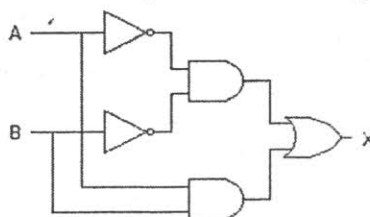
Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

1. OR gate and----- will form the NOR gate?
A) OR gate
B) NAND gate
C) NOT gate
D) AND gate
2. Decimal equivalent of binary number 10101 is
A) 21
B) 12
C) 22
D) 31
3. A three variable karnaugh map has _____
A) Eight cells
B) Three cells
C) Sixteen Cells
D) Four cells
4. Which of the following is a SOP expression?
A) $(A+B)(A+C)$
B) $AB+AC$
C) Both a and b
D) $(A+B)(C+D)$
5. Which of the following logic expressions represents the logic diagram shown?



- A) $X=AB'+A'B$
B) $X=(AB)'+AB$
C) $X=(AB)'+A'B'$
D) $X=A'B'+AB$
6. The decimal equivalent of the binary number 1011.011_2 is
A) 11.375_{10}
B) 10.123_{10}
C) 11.175_{10}
D) 9.23_{10}
7. Which of the following is not considered for forming groups in K-map?
A) Rolling
B) Diagonal
C) Vertical
D) Horizontal
8. The output of an exclusive-NOR gate is 1. Which input combination is correct?
A) $A = 1, B = 0$
B) $A = 0, B = 1$
C) $A = 0, B = 0$
D) none of these
9. How is JK flip-flop made to toggle
A) $J = 0, K = 0$
B) $J = 1, K = 0$
C) $J = 1, K = 1$
D) $J = 0, K = 1$
10. The characteristic equation for S-R flip flop is
A) $\bar{S} + \bar{R}Q$
B) $\bar{S} + RQ$
C) $S + RQ$
D) $S + \bar{R}Q$

11. Full subtractor is used to perform subtraction of _____
 A) 2 bits B) 8 bits
 C) 4 bits D) 3 bits
12. The full form of SR is
 A) System rated B) Set reset
 C) Set ready D) None of these
13. A basic S-R flip-flop can be constructed by cross-coupling of which logic gates?
 A) AND or OR gates B) XOR or XNOR gates
 C) NOR or NAND gates D) AND or NOR gates
14. Latch is a device with _____ stable states
 A) One B) Three
 C) Infinite D) Two
15. The two input MUX would have _____
 A) 1 select line B) 2 select lines
 C) 4 select lines D) 3 select lines
16. Which of the following represents a number of output lines for a decoder with 4 input lines?
 A) 15 B) 16
 C) 17 D) 18
17. The group of bits 11001 is serially shifted (right-most bit first) into a 5-bit parallel output shift register with an initial state 01110. After three clock pulses, the register contains _____.
 A) 01110 B) 00001
 C) 00101 D) 10101
18. A modulus-10 counter must have _____
 A) 4 flip-flops B) 5 flip-flops
 C) 10 flip-flops D) 3 flip-flops
19. In which one of the following counters, the flip flops are not clocked simultaneously?
 A) Synchronous counter B) Asynchronous counter/ripple counter
 C) Both A and B D) None of these
20. A shift register is a digital circuit that _____
 A) Stores data B) Shifts the data from left to right
 C) Shifts the data from right to left D) All the above

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1. a)	Convert the following functions into canonical form.				
	i) $f(w, x, y, z) = \bar{w}x + y\bar{z}$ (SOP)				
	ii) $f(a, b, c, d) = (a + \bar{b} + c)(\bar{a} + d)$ (POS)	6	L*3	1	1
b)	Using K-map and simplify the following function				
	i) $f(a, b, c) = \sum m(0, 1, 2, 3, 7)$				
	ii) $f(a, b, c, d) = \sum m(0, 1, 3, 4, 5, 6, 9, 11, 12, 13, 14)$	6	L3	2	1
c)	Define				
	i) Minterm				
	ii) Maxterm				
	iii) Canonical SOP				
	iv) Canonical POS	4	L2	1	1
2. a)	Design the combinational logic circuit which takes two, 2 bit numbers as its inputs and generates an output to indicate when the sum of the two numbers is odd.	6	L3	2	1
b)	i) $12.6532_{(10)} = (?)_{16}$				
	ii) $CF.52_{(16)} = (?)_{10}$				
	iii) $1376.185_{(10)} = (?)_2$	6	L2	1	1
c)	Perform the binary subtraction using 2's complement method.				
	i) $11010010_2 - 01011110_2$				
	ii) $00110_2 - 11010_2$	4	L3	2	1

3. a) Design a combinational logic circuit to generate an output whenever a majority of four inputs is logic 1 and output function is not specified whenever the number of 0's and 1's is equal in the inputs. However, the output is logic zero for the remaining conditions.
- b) Using the QM method simplify the following function
- $$f(a, b, c, d) = \sum m(2, 6, 8, 9, 10, 11, 14, 15)$$

8	L3	2	1
8	L3	2	1

Unit – II

4. a) Implement $f(a, b, c) = \sum m(1, 4, 5, 7)$ using
- 8:1 MUX
 - 4:1 MUX
- b) With logic diagram and function table explain the operation of Set-Reset latch made with NOR gates.
- c) Convert JK flip flop to T flip flop.
5. a) Explain the working of D flip flop with relevant logic diagram and truth table.
- b) Implement the given function using 3 to 8 decoder with minimum number of gate inputs.
- $F1(a, b, c) = \sum m(0, 1, 3)$
 - $F2(x, y, z) = \sum m(0, 2, 3, 5, 7)$
 - $F3(p, q, r) = \pi M(1, 2, 3, 5, 6, 7)$
- c) Explain the working of 8 to 3 line Encoder
6. a) What are the characteristic equations of Flip flop? Obtain the characteristic equation for
- JK Flip flop
 - D Flip Flop
- b) Explain the working of a 4-bit binary parallel adder/subtractor using XOR gates with a relevant block diagram and with an example
- c) Design the full adder using 4:1 multiplexer.

6	L3	3	1
6	L2	4	1
4	L3	4	1
6	L2	4	1
6	L3	3	1
4	L2	3	1
6	L3	4	1
6	L2	3	1
4	L2	3	1

Unit – III

7. a) Design a synchronous counter to sequence $0 \rightarrow 4 \rightarrow 1 \rightarrow 2 \rightarrow 6 \rightarrow 0$ using positive edge triggered D flip flops with minimal combinational gating.
- b) What is a Ripple counter? Sketch the logic diagram of a 3-bit ripple-up counter using asynchronous positive edge triggered T flip flops. Explain its working with the sequence table and output waveforms.
8. a) Design a synchronous counter to sequence $0 \rightarrow 2 \rightarrow 3 \rightarrow 6 \rightarrow 5 \rightarrow 1 \rightarrow 0$ using positive edge triggered T flip flops with minimal combinational gating.
- b) What is a shift register? With the help of a suitable example explain the following operations:
- SIPO
 - PISO

8	L3	5	1
8	L2	5	1
8	L3	5	1
8	L2	5	1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
